

ESP-ROLL Firmware Requirements - Updated

Overview

This document outlines the comprehensive firmware requirements for the ESP-ROLL spherical self-balancing robot platform, including all new sensor integrations, wireless charging capabilities, and enhanced safety/security features.

Hardware

Core Platform

- ESP32 microcontroller with dual-core processing
- • Wireless charging receiver module - ADDED
- • Enhanced power management system
- • 3D-printed chassis with clear ornament shell
- • Compact form factor for indoor and outdoor use

Motors & Movement

- Dual geared motors (400RPM micro metal)
- • DRV8833 motor driver
- • Self-balancing mechanism without external sensors or wheels
- • Spherical design for smooth rolling over various surfaces

Sensors

New Sensor Array - ADDED

- High-resolution camera module - ADDED
- • Digital microphone array - ADDED
- • Temperature sensor (ambient & internal) - ADDED
- • Infrared (IR) sensor for night vision - ADDED
- • Ultrasonic distance sensor - ADDED

Existing Sensors

- XIAO ESP32S3 Sense Camera module
- • Integrated IMU/gyroscope for balance control
- • Battery monitoring system

● AI/Connectivity

Enhanced AI Features - ADDED

- On-device computer vision processing - ADDED
- • Voice recognition and response - ADDED
- • Environmental monitoring and alerts - ADDED
- • Autonomous navigation with obstacle avoidance - ADDED

Wireless Connectivity

- WiFi control and live FPV video streaming
- • Real-time data transmission
- • Remote monitoring capabilities

Web/Control

User Interface

- Web-based control panel
- • Live video streaming interface
- • Real-time sensor data dashboard
- • Mobile app compatibility - ADDED

Control Systems

- Manual remote control
- • Autonomous operation modes - ADDED
- • Programmable behavior patterns

Power

Wireless Charging System - ADDED

- Qi-compatible wireless charging pad - ADDED
- • Automatic docking and charging - ADDED
- • Charging status indicators - ADDED

Battery Management

- 1S battery power system with integrated charging
- • Enhanced battery monitoring
- • Low-power sleep modes
- • Extended battery life optimization - ADDED

Safety/Security

Enhanced Security Features - ADDED

- Encrypted data transmission - **ADDED**
- • Secure authentication protocols - ADDED
- • Privacy protection for camera/microphone - ADDED
- • Intrusion detection alerts - ADDED

Safety Systems

- LED ring light system for visibility
- • Active buzzer for alerts and notifications
- • Emergency stop functionality - ADDED
- • Safe operating temperature monitoring - ADDED

Software & OTA

Over-the-Air Updates - ADDED

- Secure firmware update mechanism - ADDED
- • Automatic update notifications - ADDED
- • Rollback capability for failed updates - ADDED

Software Features

- Custom PCB with integrated peripherals
- • Modular firmware architecture
- • Machine learning model updates - ADDED
- • Cloud connectivity for advanced features - ADDED
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ESP-ROLL Project Overview

This firmware uses and extends the ESP-ROLL project

(<https://www.instructables.com/ESP-ROLL-Build-a-Spherical-Self-balancing-Robot-Wi/>), a spherical self-balancing robot platform developed for ESP32 microcontrollers.

Base Platform: ESP-ROLL Spherical Self-Balancing Robot

The ESP-ROLL is a small, home-built spherical robot that operates inside a transparent ornament ball. Key characteristics of the base platform include:

- Spherical design that rolls smoothly over various surfaces (tile, carpet, grass, sand)
- • Self-balancing mechanism without external sensors or wheels
- • ESP32 camera integration for WiFi control and live FPV video
- • 1S battery power system with integrated charging
- • 3D-printed chassis with clear ornament shell
- • Compact form factor suitable for indoor and outdoor use

Core Components:

- XIAO ESP32S3 Sense Camera module
- • Dual geared motors (400RPM micro metal)
- • DRV8833 motor driver
- • Custom PCB with integrated peripherals
- • LED ring light system
- • Active buzzer for alerts
- • Battery monitoring system

Additional Capabilities and Extensions

This firmware implementation extends the base platform with the following enhancements: